

Experiment 28

Aim

To find the boundary of an image using a convolution kernel in OpenCV.

Procedure

1. Import the required libraries, OpenCV and NumPy.
2. Read the input image.
3. Convert the image into grayscale.
4. Define a convolution kernel for detecting boundaries.
5. Apply the kernel to the grayscale image using cv2.filter2D().
6. Display the original and boundary-detected images.
7. Save the boundary image.

Code:

```
import cv2
import numpy as np
img = cv2.imread("image.jpg")
if img is None:
    print("Error: Image not found!")
    exit()
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
kernel = np.array([
    [-1, -1, -1],
    [-1, 8, -1],
    [-1, -1, -1]
])
boundary = cv2.filter2D(gray, -1, kernel)
cv2.imshow("Original Image", img)
cv2.imshow("Boundary Image", boundary)
```

```
cv2.imwrite("boundary_image.jpg", boundary)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Sample Input:



Output:



Result

The boundary of the given image was successfully detected using a convolution kernel. The detected boundaries were displayed and saved as boundary_image.jpg.